Get Started with facet

Introduction

The `facet` package provides a unified framework for Generalizability Theory (G-Theory). In this quick start, we will walk through a standard analysis: conducting a G-study, and then running a D-study.

1. Load the Package and Data

First, load `facet` and one of the built-in datasets, `brennan`.

```
library(facet)
data(brennan)
head(brennan)
#>   Task Person Rater Score y
#> 1    1      1     1     5 5
#> 2    1      2     1     9 9
#> 3    1      3     1     3 3
#> 4    1      4     1     7 7
#> 5    1      5     1     9 9
#> 6    1      6     1     3 3
```

2. Conduct a G-Study

Use the `gstudy()` function to estimate variance components. By default, `facet` uses `lme4` (Restricted Maximum Likelihood) under the hood.

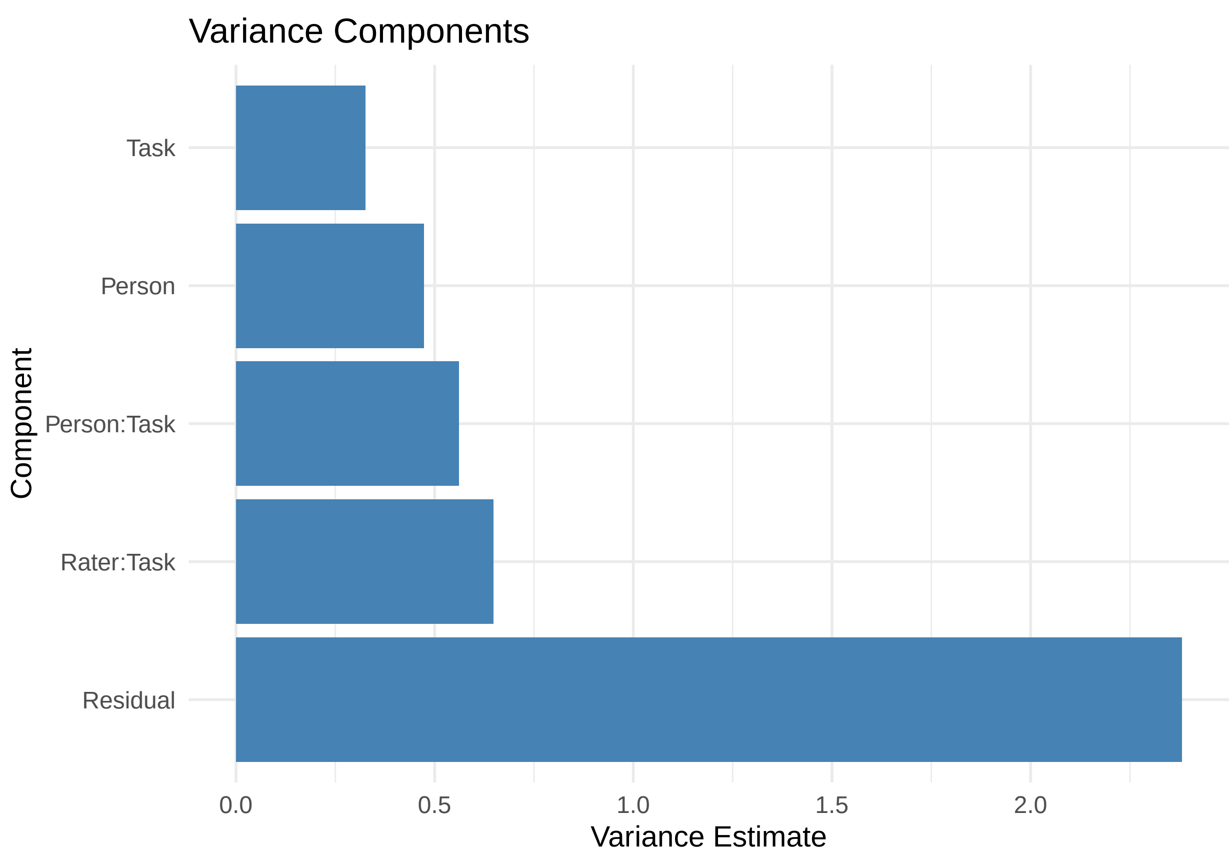
```
g_obj <- gstudy(Score ~ (1 | Person) + (1 | Task) + (1 | Rater:Task) +
  (1 | Person:Task),
  data = brennan)

# View the variance components
print(g_obj)
#> Generalizability Study (G-Study)
#> ================================
#>
#> Backend: lme4
#> Formula: Score ~ (1 | Person) + (1 | Task) + (1 | Rater:Task) + (1 | Person:Task)
#> Number of observations: 120
#> Multivariate: No
#>
#> Object of measurement: Person
#> Facets: Person, Task, Rater
#>
#> Sample Sizes:
#> # A tibble: 6 x 3
#>   effect              type            n
#>   <chr>               <chr>       <dbl>
#> 1 Person              main           10
```

```
#> 2 Task                 main              3
#> 3 Rater:Task           interaction      12
#> 4 Person:Task          interaction      30
#> 5 Person:Rater (res)   residual        120
#> 6 Rater per Task       nested            4
#>
#> Nested details:
#>   Rater per Task: 3 Tasks, 4.0 Raters per Task
#>
#> Variance Components:
#> # A tibble: 5 x 3
#>   component   estimate   pct
#>   <chr>          <dbl> <dbl>
#> 1 Person         0.473 10.8
#> 2 Task           0.325  7.42
#> 3 Rater:Task     0.647 14.8
#> 4 Person:Task    0.56  12.8
#> 5 Residual       2.38  54.3
```

3. Visualize Variance Components

Plotting the random effects helps identify which facets contribute the most variance.

```r
plot(g_obj)
```



4. Conduct a D-Study

A Decision (D) study calculates the dependability of your measurement design. You can project how changing the sample sizes of your facets impacts reliability.

```
# Project a design with 3 Tasks and 4 Raters
d_obj <- dstudy(g_obj, n = list(Task = 3, Rater = 4))

print(d_obj)
#> Decision Study (D-Study)
#> ========================
#>
#> Based on G-Study with lme4 backend
#> Object of measurement: Person
#> Universe components: Person
#> Error components for relative error (sigma2_delta): Person:Task, Person:Rater (Residual)
#> Error components for absolute error (sigma2_delta_abs): Task, Rater:Task, Person:Task, Person:Rater
#>
#> Sample Sizes:
#>  Task: 3
#>  Rater: 4
#>
#> Variance Components:
#> # A tibble: 5 x 6
#>   component   dim   var_unscaled pct_unscaled var_scaled pct_scaled
#>   <chr>       <chr>        <dbl>        <dbl>      <dbl>      <dbl>
#> 1 Person      Score        0.473        10.8       0.473      33.4
#> 2 Task        Score        0.325         7.41      0.108       7.65
#> 3 Rater:Task  Score        0.647        14.8       0.054       3.80
#> 4 Person:Task Score        0.56         12.8       0.187      13.2
#> 5 Residual    Score        2.38         54.3       0.595      42.0
#>
#> Coefficients:
#>    dim   uni sigma2_delta sigma2_delta_abs     g   phi
#>  Score 0.473        0.782            0.944 0.377 0.334
```

Next Steps

For comprehensive documentation, see `vignette("introduction")`, which covers:

- Historical foundations of generalizability theory
- All three estimation backends (MOM, REML, Bayesian)
- Univariate and multivariate G-studies and D-studies
- Value Added Ratios and composite score optimization
- Advanced topics, reporting guidelines, and exercises